

Name: _____ Date: _____

Period: _____

Part 1 — Evaluate

Evaluate each expression. Give the exact answer in radians.

1. $\arcsin\left(\frac{\sqrt{2}}{2}\right) = \underline{\hspace{2cm}}$

2. $\arccos\left(-\frac{1}{2}\right) = \underline{\hspace{2cm}}$

3. $\arctan\left(\frac{\sqrt{3}}{3}\right) = \underline{\hspace{2cm}}$

4. $\sin^{-1}(0) = \underline{\hspace{2cm}}$

5. $\cos^{-1}(1) = \underline{\hspace{2cm}}$

6. $\tan^{-1}\left(-\frac{\sqrt{3}}{3}\right) = \underline{\hspace{2cm}}$

Part 2 — Domain and Range

7. State the domain and range of each function.

Function	Domain	Range
$y = \arcsin x$	<u> </u>	<u> </u>
$y = \arccos x$	<u> </u>	<u> </u>
$y = \arctan x$	<u> </u>	<u> </u>

Part 3 — AP-Style Multiple Choice8. Which of the following is the value of $\arccos\left(-\frac{\sqrt{2}}{2}\right)$?

(A) $-\frac{3\pi}{4}$

(B) $-\frac{\pi}{4}$

(C) $\frac{\pi}{4}$

(D) $\frac{3\pi}{4}$

(E) $\frac{5\pi}{4}$

9. Let $f(x) = \arctan(x)$. Which statement about f is true?(A) The range of f is all real numbers.(B) $f(x)$ is undefined for $x < -1$.(C) The range of f is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.(D) $f(-x) = -f(x)$ is false for all $x \neq 0$.**Extension** optional

Extension — Composition of Trig and Inverse Trig

10. Simplify $\sin(\arctan x)$ for $x > 0$. Express your answer without any trig or inverse trig functions.

Preview Lesson 3.10: Consider the equation $\cos \theta = \frac{1}{2}$. How many solutions exist on $[0, 2\pi)$? List them.

How would you use arccos to find one solution? We will explore this in the next lesson.
